

Weiching Chen

weichingchen73@gmail.com | (857) 800-3793 | [LinkedIn](#) | U.S. Permanent Resident

Personal Website: <https://weiching-chen.github.io/weiching-chen>

EDUCATION

Cornell University - M.S. in Design Technology

Expected 2026

Relevant Coursework: Robotics, Reinforcement Learning, Human Robot Interaction (HRI), Extended Reality (XR) Computing, Embedded Systems, Robotic Fabrication.

National Taiwan University - B.S. in Electrical Engineering

2024

Relevant Coursework: Control Systems, Robotics, Embedded Systems, Full-stack Web Development, Machine Learning, Algorithms, Data Structures.

SKILLS

Robotics & AI: ROS/ROS 2, PyTorch, NVIDIA Isaac Lab/Sim, MoveIt 2, OpenCV, NVIDIA Jetson.

Robotic Platforms: ABB IRB 120/6700, Unitree Go2/G1, Universal Robots UR5e/UR10e, PhantomX, TurtleBot 4, HoloLens 2, Meta Quest.

Programming & Systems: Python, C/C++, Docker, Linux (Ubuntu), JavaScript (React.js), C# (Unity), PostgreSQL, STM32, Raspberry Pi, ESP32, MQTT, WebSockets, Qt5, AWS.

Design & Fabrication: Rhino (Grasshopper/Kangaroo), SolidWorks, G-code (Prusa), Adobe Creative Suite.

EXPERIENCE

Nexuni

Taiwan

Co-Founder / Founding Member of Technical Staff

June 2019 – Present

Co-founded and scaled a 40+ person startup, leading and executing key projects in full-stack infrastructure, autonomous robotics, and Human-Robot Interaction (HRI) systems.

- Engineered a hybrid full-stack architecture featuring a Qt5 payment interface and AWS cloud backend for automation store (DoDay) operations to ensure 100% service continuity over 6 years, eliminating revenue loss and scaling the solution to 50+ commercial parking locations.
- Deployed a ROS 2-based interface and WebRTC/RTSP dashboard for Unitree Go2 to enable low-latency monitoring and real-time point-and-click path planning, achieving successful commercialization as an autonomous patrolling system in major Singapore shopping malls.
- Developed a teleoperation system integrating Unitree G1 humanoid robots with Meta Quest to facilitate immersive hand-tracking control, leveraging custom gesture recognition for high-precision remote manipulation.
- Developed an RL training pipeline and policy in NVIDIA Isaac Sim for quadruped robots to master complex vertical stair climbing, significantly improving robustness and navigation performance on high-difficulty terrains.
- Streamlined internal development workflows by creating proprietary pip (Python) and npm (JavaScript) packages to standardize UI components and backend logic across online ordering and parking management platforms.

Cornell University

New York, NY

Graduate Research Assistant

Sep 2024 – Present

- Created a standalone ROS 2 control package from scratch for the PhantomX arm to provide the only available ROS 2 support for this platform, achieving real-time teleoperation via Unity and Meta Quest.
- Designed collision-aware toolpaths for large-scale timber processing on an ABB IRB 6700 to automate industrial wood cutting, translating complex geometric data into executable RAPID code via RobotStudio.
- Implemented a low-latency robotic control interface using ABB EGM (Externally Guided Motion) to facilitate high-speed Python API control, integrating HoloLens 2 hand-tracking for real-time human-robot collaborative assembly on the ABB IRB 120.
- Optimized non-planar 3D printing for the Prusa MK4 by developing a Python-based G-code compiler to bypass standard slicer limitations, achieving support-free fabrication of complex geometries.

HONORS & AWARDS

Google Science Fair Global Finalist (Top 20 Worldwide, 2019); **Canada Worldwide Science Fair Gold Medal** (2018); **Taiwan International Science Fair Silver Award** (2018); **LINE FRESH Hackathon 2nd Place** (Competed against over 1,000 teams, 2020).